HireMind: Efficient CV–Job Fit Classification through Distillation and Quantization

Rawan Khojah **Hams Aljohani**
Effat University
CS4085 Deep Learning

1 Motivation and related work

HireMind is our Senior recruitment platform. We investigate a compact CV–job fit classifier that could later provide one piece of evidence to its scoring workflow; it does not replace skill verification or recruiter decisions, and platform integration is future work. A small local model could reduce reliance on external API calls for matching and keep this step inside an organization, but we only claim what we measure: quality, storage, latency and memory.

Gan et al. (2024) motivate automated resume screening. BERT (Devlin et al., 2019) provides the architecture, and Turc et al. (2019) show that pretrained compact students benefit from distillation: a 6-layer student's GLUE development meta-score rises from 82.8 to 84.4. Hinton et al. (2015) introduce temperature-based distillation; on MNIST, a distilled small network makes 74 errors instead of 146. Q8BERT (Zafrir et al., 2019) shows that 8-bit quantization-aware training keeps MRPC F1 near FP32 (89.56 vs. 90.0), while post-training dynamic quantization loses more (87.88). **Gap:** these five sources study distillation and quantization separately; none asks whether a distillation gain in a 4.4M-parameter student survives *post-training* 8-bit quantization that also covers the embeddings, which hold about 90% of BERT-Tiny's weights (3.97M of 4.4M parameters). This matters: if quantization erases the gain, the cheaper hard-label model would be enough. We study this combination empirically on CV–job pairs, without claiming a new algorithm.

2 Question and proposed approach

Research question: Can teacher-guided training improve compact CV–job fit classification, and can post-training quantization cut storage while keeping that improvement?

Input is a CV and job description; output is *No Fit*, *Potential Fit* or *Good Fit*. The student is Google's pretrained BERT-Tiny, `bert_uncased_L-2_H-128_A-2`, with a three-class head. A fine-tuned `bert-base-uncased` teacher (same three-class head) supplies logits.

M0: student trained with hard-label cross-entropy. **M1:** same architecture and initialization, trained with

$$L = 0.5L_{CE} + 0.5\,T^2\,\mathrm{KL}\big(\sigma(z_T/T)\,\|\,\sigma(z_S/T)\big), \tag{1}$$

where z are logits and σ is softmax; the T^2 factor is implemented. **M2:** a CPU copy of M1 converted with PyTorch `quantize_dynamic`: Linear layers use dynamic INT8 (`default_dynamic_qconfig`) and embeddings 8-bit weight-only (`float_qparams_weight_only_qconfig`); layer normalization stays FP32. M2 needs no extra training. Distillation is Part 1; quantizing its result is Part 2. All models share preprocessing and a 256-token limit; truncation frequency is reported.

3 Hypotheses

Let F_i be mean test macro-F1 (0–100) over three seeds and S_i serialized size. **H1:** $F_1 - F_0 \geq 1.0$ point. **H2:** $S_2 \leq 0.40S_1$, $F_1 - F_2 \leq 1.0$ point, and $F_2 - F_0 \geq 0.5(F_1 - F_0)$. Targets are exploratory: 60% is motivated by the embedding share, not by a pilot. H2 is fixed now and is not changed after the pilot. If H1 fails, H2's retention part is not confirmatory and only compression is reported.

4 Data and fair experiments

We use the instructor-approved resume–job-description fit dataset on Hugging Face,[1] pinned to revision `08978e2`: 8,000 pairs (6,241 train, 1,759 test), 643 CVs, 351 jobs, labels 4,000/2,000/2,000.

[1] https://huggingface.co/datasets/cnamuangtoun/resume-job-description-fit

CV and job IDs are SHA-256 hashes of whitespace-normalized text. Rows of the six conflicting pairs (12 rows) are dropped from whichever split holds them; for the one exact duplicate, a copy in training is dropped if the pair also appears in test, leaving 7,987 unique pairs. About 20% of training jobs, grouped by job ID, form validation (seed 42), so no validation or test job appears in training; the test file is otherwise untouched. The audit script outputs the final train/validation/test counts, which are published with split indices and file hashes on GitHub before training; CV text is not redistributed. The official split shares 476 CVs but no jobs, so we measure *unseen jobs with possibly familiar CVs*, not new candidates.

Label validity. Provenance, licence and labeling method are undocumented. Before training, both authors independently label 90 training pairs, sampled randomly within each class (30 per class, seed 42), using a written rubric on skills, experience and role match. We report Cohen's κ, agreement with dataset labels and per-class disagreements. This is a limited audit, not validation of all labels; test scores mean agreement with benchmark labels, not hiring validity.

Training. AdamW, effective batch 16, at most three epochs, equal update caps. M0 tries four learning rates; M1 tries two learning rates $\times\, T \in \{2, 4\}$, so both get four validation trials. Selected settings are rerun as three independent seeds (42–44) from the same pretrained start; we report all three paired differences and their mean $\pm$ s.d., without significance claims. The teacher is trained once on training data, selected on validation, and its cached logits are reused for every M1 run. All choices are frozen before the test set is opened.

Measurement. Macro-F1, accuracy, per-class scores and confusion matrices. The primary compression measure is serialized size: bytes of the model `state_dict` saved with `torch.save`, excluding tokenizer and config, identically for M1 and M2. Latency and memory are separate empirical outcomes, not assumed consequences of quantization. Latency uses one pinned PyTorch version on one stated CPU, one thread, batch size 1, 256 tokens, 30 warm-up and 200 timed calls (median, p95), reporting model-only and tokenization-inclusive times separately, plus peak memory. MACs describe the operation graph, not realized speed.

5 Feasibility and team plan

No model has been trained. A Colab pilot on a small subset must pass: (1) Linear and Embedding modules convert; (2) M2 saves and reloads, with embedding weights still stored in 8-bit; (3) outputs are finite; (4) M2 agrees with M1 on at least 95% of validation predictions. We also record serialized size, latency and memory. Our two-person team is approved by the instructor. Rawan leads data auditing and measurement; Hams leads training and export; both verify and write. Track: Technical Report. GitHub and Overleaf projects are shared with the instructor. Weeks (to be confirmed with the calendar): 10 labels and split; 11 pilot, teacher, M0; 12 M1; 13 M2 and profiling; 14 final evaluation; 15 report, presentation, poster.

6 Risks and fallback

If embedding quantization fails the pilot, H2 is reported as untestable for the planned M2; a Linear-only variant is then evaluated and reported separately as exploratory, with no success target. If the teacher does not beat M0 on validation, M1 still runs as planned and is reported; class-weighted training would be a separate, labelled exploratory run, never a replacement for M1. Memory limits: gradient accumulation. Label noise and truncation are reported as limitations. Predictions will not support autonomous hiring or fairness claims.

7 AI use

ChatGPT and Claude assisted literature analysis, dataset research and auditing, experiment planning, workspace setup and drafting. The authors checked the content and references and are responsible for the submission. No experimental results were generated or claimed.

References

Chengguang Gan, Qinghao Zhang, and Tatsunori Mori. 2024. Application of LLM Agents in Recruitment: A Novel Framework for Automated Resume Screening. *Journal of Information Processing*, 32:881–893. DOI: 10.2197/ipsjjip.32.881.

Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. 2019. BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding. *NAACL-HLT*, pages 4171–4186. https://aclanthology.org/N19-1423/.

Iulia Turc, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. 2019. Well-Read Students Learn Better: On the Importance of Pre-training Compact Models. *arXiv:1908.08962v2*. https://arxiv.org/abs/1908.08962v2.

Geoffrey Hinton, Oriol Vinyals, and Jeff Dean. 2015. Distilling the Knowledge in a Neural Network. *arXiv:1503.02531*. https://arxiv.org/abs/1503.02531.

Ofir Zafrir, Guy Boudoukh, Peter Izsak, and Moshe Wasserblat. 2019. Q8BERT: Quantized 8Bit BERT. *EMC2 workshop*. DOI: 10.1109/EMC2-NIPS53020.2019.00016. https://arxiv.org/abs/1910.06188v2.

A AI review report

Review generated by ChatGPT on 18 October 2026 in a new conversation, using the course AI Reviewer prompt and the complete proposal text. Web-citation labels inserted by the chat interface were removed; the review content is otherwise reproduced unchanged.

CS4085 Deep Learning — Formative Proposal Review

Project: HireMind: Efficient CV–Job Fit Classification through Distillation and Quantization

Review standard: Strict, rubric-based assessment of the proposal as written, not an official course grade. I am evaluating research design, technical correctness, reproducibility, feasibility, and the strength of the evidence promised by the plan. I am not requiring full model training at the proposal stage.

1. Overall assessment

Provisional formative score

83/100

READY TO SUBMIT*

*Under the stated numerical threshold, provided the technical and experimental details below are corrected or verified before submission.

The proposal is unusually well specified for a project that has not yet trained a model. It defines all three models, connects the two contributions, sets numerical hypotheses, distinguishes validation from test data, and plans reproducible measurements.

The main weakness is not missing structure. It is the gap between the intended quantization experiment and what the proposed implementation and pilot can reliably establish. The literature gap also needs a more careful qualification, and the experimental design needs stronger safeguards against candidate-level and dataset-level leakage.

My strict-reviewer verdict: a strong proposal with several important technical and methodological details to resolve. It should not be treated as a completed research result. The absence of trained models is acceptable at this stage, and you explicitly disclose that no results exist.

2. Category breakdown

Rubric category	Score	Reviewer assessment
Motivation	9/10	Practical motivation, bounded claims, and a clear efficiency rationale.
Related work (at least 5 papers)	8/10	Five sources are cited, but the synthesis and novelty claim need stronger verification.
Research question and gap	9/10	Specific, testable question and a clear proposed gap.
Deep learning contribution	13/15	M0/M1 are well defined; teacher details and the interpretation of the gain need tightening.
Efficiency contribution	11/15	M2 is specified, but quantization support, implementation details, and the compression target need validation.
Two hypotheses	9/10	Numerical, connected, and appropriately conditional.
Dataset and fair experiments	7/10	Strong split discipline, but repeated-CV overlap, label validity, and the tuning protocol deserve more scrutiny.
Feasibility and team plan	8/10	Sensible pilot and division of work; the proposed quantization path remains a technical risk.
Risks and fallback	4/5	Honest fallbacks; the Linear-only fallback changes the efficiency method and cannot establish the original H2.
Writing, format, AI disclosure	5/5	Clear structure, measurable claims, and transparent AI-use disclosure.
Total	83/100	Strong proposal; revisions recommended.

These are formative estimates, not an official grade. The score reflects the written plan, not experimental performance.

Mandatory-rule audit M0, M1, and M2 defined

M0 is the hard-label student baseline, M1 adds teacher-guided distillation, and M2 quantizes M1. Their relationship is explicit.

H1 compares M1 against M0

$F_1 - F_0 \geq 1.0$ macro-F1 point. This directly tests the deep learning contribution.

H2 compares M2 against M1 and assesses M2 against M0

H2 tests size reduction, the loss relative to M1, and retention of at least half the M1-versus-M0 gain. This is a meaningful connected efficiency contribution.

Numerical targets present

The proposal gives explicit thresholds for both hypotheses and a 95% validation-prediction agreement criterion for the pilot.

No stated test-set tuning

Hyperparameter selection is assigned to validation, and test access is deferred until choices are

frozen. This is the correct intended protocol.

One important qualification: these checks confirm that the proposal states the required design. The implementation, data audit, and eventual experiment logs must demonstrate that the design is followed.

3. Three main improvements

1. Prove that the planned quantization works on the actual model This is the most important technical issue. The proposal specifies dynamic INT8 quantization for Linear layers and weight-only INT8 quantization for Embedding layers. Those are not interchangeable operations, and successful conversion of one does not establish that the full Hugging Face model will convert and run correctly.

PyTorch provides dedicated embedding quantization support and configuration requirements. Verify compatibility with your exact model architecture, installed PyTorch version, and save/reload path.

Your existing pilot is a good start. Strengthen its evidence by confirming:

- The actual embedding weight dtype and representation after conversion, rather than relying only on a successful conversion call.
- Successful inference after saving and reloading the entire model.
- The actual serialized-size ratio S_2/S_1.
- Finite outputs and macro-F1 on a small validation subset, alongside the prediction-agreement check.
- The exact modules quantized and those left in FP32.

Why this matters: If embeddings are not quantized as intended, M2 may miss its compression target even if Linear quantization works. A passing conversion test is not the same as a passing efficiency experiment.

2. Make the literature gap more defensible The proposal cites five relevant sources, and the combination of distillation and quantization is a plausible experimental question. However, the claim that no prior work examines the combination is stronger than the evidence provided.

Q8BERT studies quantization-aware training, whereas your proposed M2 uses post-training quantization. That difference helps motivate your experiment, but it does not by itself establish that the overall combination is novel. The compact-model distillation literature also includes work beyond the one result quoted from Turc et al.

Improve the research argument by verifying whether prior work has studied:

- Distilled compact BERT models followed by post-training quantization.
- Quantization of embedding weights in compact Transformer models.
- The interaction between distillation gains and post-training quantization, even on tasks other than recruitment.

Then qualify the gap to match what you can substantiate. You do not need to claim a new algorithm: a well-controlled, task-specific empirical study is a reasonable course contribution.

3. Strengthen the data and evaluation argument The split protocol is more thoughtful than a random row-level split, but the proposal reports 476 shared CVs between the official training and test portions. It correctly acknowledges that the experiment therefore measures performance on unseen jobs with potentially familiar CVs.

That is an important limitation, not a trivial detail. A classifier can potentially exploit resume-specific patterns when a CV appears in both training and test data. This may affect how you interpret the measured fit-classification performance.

Before training, ensure the audit clearly documents:

- Exact CV and job overlap after deduplication and conflict removal.
- The final train, validation, and test counts after every filtering operation.
- Whether your intended claim is generalization to unseen jobs, unseen candidates, or both.
- How the reported results might change if candidate-level separation were used.

The existing protocol can support a narrowly scoped study of unseen jobs. It cannot, on its own, establish generalization to unseen candidates. Keep that distinction explicit in your report.

4. Major and minor issues

Major issues These are the items I would expect a strict instructor to challenge.

Major 1The full-model quantization path is not yet demonstrated.

The proposed API and embedding configuration must work together on the exact BERT-Tiny implementation. Some Transformer operations may remain floating point, and converting weights does not guarantee faster CPU inference. PyTorch's documented dynamic quantization support is module-specific, so compatibility must be established rather than assumed.

Required before submission: a successful small pilot, or a clearly documented technical uncertainty and fallback. Full training is not required.

Major 2The novelty claim is not sufficiently established.

Five references meet the minimum numerically, but the claim that prior studies do not examine this combination needs a broader literature check. The papers demonstrate related techniques; they do not prove the absence of earlier combined studies.

Major 3The evaluation scope is narrower than a general CV–job fit claim.

Your reported CV overlap is acknowledged, which is good. However, the benchmark cannot establish generalization to unseen candidates. The final report must consistently describe the intended unseen-job evaluation and avoid stronger claims.

Major 4The compression target needs an empirical justification.

The 0.40 size ratio is a legitimate, falsifiable target, but the 90% embedding parameter share does not directly imply a 60% reduction in total serialized size. Parameter count, storage dtype, quantization metadata, and unquantized components all matter.

The target is not inherently unreasonable. Present it as a hypothesis motivated by the architecture, not as a result implied by the embedding share.

Minor issues

- Teacher selection: Specify the validation metric used to select the teacher checkpoint and the exact stopping or checkpoint-selection rule.
- Distillation reproducibility: State the temperature and loss-weight search procedure completely, including how the best configuration is chosen and whether the teacher's logits are computed in evaluation mode.
- Three-seed uncertainty: Three seeds are acceptable for a small course project, but the mean and standard deviation are descriptive, not strong evidence of statistical significance. Your proposal already acknowledges this.
- Prediction-agreement pilot criterion: At least 95% agreement with M1 is a useful diagnostic, but it is not evidence that M2 preserves task performance. Keep the separate macro-F1 hypothesis.
- Latency and memory: Specify the CPU model, operating system, software versions, and memory-measurement method. Distinguish model weight storage from runtime peak memory.
- Fallback interpretation: A Linear-only quantization variant can be useful exploratory evidence, but it is a different M2. It must not be presented as a successful test of the original embedding-plus-Linear quantization hypothesis.
- Label audit: Independently labeling 90 training pairs is useful. However, Cohen's kappa measures annotator agreement, not whether the dataset's labels reflect actual hiring outcomes.
- Dataset provenance and license: The linked dataset is listed as containing 8,000 rows, and its repository exposes the pinned revision. The dataset page does not provide a dataset card, so provenance and labeling details remain unresolved.

Verify the license and permitted use of the underlying CV text before publication or redistribution.

- AI disclosure: This is appropriately transparent. Confirm that it meets Effat University's applicable disclosure requirements; I cannot determine course-specific compliance from the proposal alone.

5. References you must verify yourselves

I checked the discoverable bibliographic records for the cited works. This is a preliminary metadata check, not a substitute for reading the original papers and verifying the reported numerical claims.

1. Devlin et al. (2019) — BERT

Status: Bibliographic record identifiable.

Verify that your description of BERT supports the specific architecture and modeling choices you use, rather than treating the original BERT paper as evidence for every property of BERT-Tiny.

ACL Anthology: original paper

1. Turc et al. (2019 preprint; ICLR 2020)

Status: Paper verified; numerical claim requires checking.

The work studies pretrained compact models and distillation. Verify the exact GLUE development score comparison (82.8 versus 84.4), including which student and evaluation setup the numbers describe. The work is also listed as an ICLR 2020 paper, so check the preferred publication citation.

arXiv: original paper

1. Hinton, Vinyals, and Dean (2015) — Distilling the Knowledge in a Neural Network

Status: Original preprint identifiable; result needs verification.

Check the cited MNIST error counts (74 versus 146) against the original experiment and ensure the comparison is described with its proper model and training context.

arXiv: original paper

1. Zafrir et al. (2019) — Q8BERT

Status: Bibliographic record identifiable; comparison requires checking.

Verify the MRPC F1 values (90.0, 89.56, and 87.88), their experimental settings, and whether the quoted results refer to the quantization procedures you describe. Q8BERT's quantization-aware training differs from your post-training approach.

arXiv: original paper

1. Gan, Zhang, and Mori (2024) — Application of LLM Agents in Recruitment

Status: Journal record verified.

The authors, journal, volume, pages, year, and DOI are consistent with the publisher's record. Read the paper to verify that it supports the specific motivation attributed to it; its LLM-agent recruitment framework is not equivalent to your compact three-class classifier.

Publisher's article page

Additional source verification Beyond the five cited papers, verify the following primary sources and technical claims:

- BERT-Tiny checkpoint: Confirm the provenance, configuration, vocabulary, and parameter count of `google/bert_uncased_L-2_H-128_A-2`. Do not assume that the original BERT paper establishes the exact 4.4M parameter count for your checkpoint.
- PyTorch quantization documentation: Check the installed version's API behavior, supported modules, embedding configuration, and serialization procedure. The current documentation is a useful starting point, not proof of compatibility with your environment. PyTorch quantization recipe.
- Dataset revision and split: Download the exact revision `08978e2`, independently run the audit, and verify every count after removing conflicting and duplicate pairs. Pinned dataset repository.
- Prior combined methods: Search the references and related work of the five papers for studies combining distillation and post-training quantization. Until that search is complete, the gap is plausible but not conclusively established.

I have not independently reproduced the five papers' experimental results, your dataset audit, or the proposed quantization behavior. None of the proposed project's outcomes should be described as established findings.

6. Three questions your instructor may ask

Question 1: What is genuinely your deep learning contribution if knowledge distillation already exists?

Be prepared to explain that the contribution is the controlled investigation of whether teacher-guided training improves a pretrained compact student on this task, using M0 as the matched hard-label baseline. Do not claim that the distillation algorithm itself is new.

Question 2: Why should quantizing the embeddings and Linear layers produce a 60% storage reduction, and how will you prove it?

Explain that 0.40 is a target to test, not a guaranteed consequence of the embedding parameter

share. Describe the pilot's conversion checks, saved-state-dict measurement, and planned comparison of M2 with M1 and M0.

Question 3: How can you trust your test results if the same CVs appear in training and test, and the dataset's labeling method is undocumented?

Explain the narrower unseen-job evaluation, the deduplication and overlap audit, and the independent 90-pair label audit. Acknowledge that these measures reduce some risks but do not establish generalization to new candidates or validate the labels as real hiring outcomes.

Final recommendation

Readiness: READY TO SUBMIT by the stated score threshold, with important caveats.

Your strongest features are the explicit M0/M1/M2 definitions, the connected H1/H2 hypotheses, the frozen numerical targets, and the separation of validation from test evaluation. Those are substantial strengths for a formative proposal.

Before submitting, prioritize these three actions:

1. Validate the complete quantization and reload path on the actual checkpoint and intended PyTorch environment.

2. Substantiate and carefully scope the literature gap, including verification of all reported paper-specific numbers.

3. Make the data limitations and evaluation scope unmistakable, particularly shared CVs, undocumented labels, and the distinction between unseen jobs and unseen candidates.

You do not need to train the full model to address these proposal-stage concerns. A small, documented pilot is sufficient to establish technical feasibility. If the pilot cannot verify the planned M2, or the literature review cannot support the claimed gap, resolve those issues before relying on the current readiness assessment.